

Review Article

Phosphorene polymeric nanocomposites for electrochemical energy storage applications

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ABSTRACT

Nanotechnological advances in 2-D nanomaterials (NMs) have recently increased their extent of usage as multifunctional NMs and their embedment in varying polymeric matrices have incorporated enhanced features for multifaceted applications. On the other hand, as a result of inherent flaws in certain 2-D NMs (carbon nanotubes (CNTs), graphene (GN), and so on), the quest for new 2-D NMs with capacity to fill the void caused by these limitations has continually escalated, thereby inducing phosphorene (P—N) development, an emerging 2-D NM, similar to GN, and an allotrope of phosphorous (PH). The incorporation of P—N within varying polymeric matrices have resulted in the formation of superior P—N@polymeric nanoarchitectures applicable in electrochemical energy storage devices (EESD) (supercapacitors, lithium ion batteries (LIBs) along with emerging technologies (sodium ion batteries, lithium–sulfur batteries, and magnesium ion batteries), electronics, photo-thermal gadgets, sensing, optoelectronics, biomedical, photovoltaic, and so on. Hence, the current elucidation offers insight into the design, development, characterization, and EESD application of P—N and P—N@polymeric nanoarchitectures. Furthermore, application in energy conversion gadgets and photo-catalysts are elucidated.

1. Introduction

Previously, 2-D NMs including graphene (GN), hexagonal boron nitride (h-BN), hydroxides, layered metallic oxides, metallic dichalcogenides, graphitic carbon nitride, carbon nanotubes (CNT), MXenes (M—X), and so on, have garnered great attention. Currently, emerging 2-D NMs similar to GN (silicene, phosphorene (P—N), germanene, siloxene, antimonene, and so on), have emanated and strongly competing with already established 2-D NMs such as GN as replacements or alternative NM [1]. Varying research results have revealed prospects of GN in enhancing features of electrochemical energy storage devices (EESDs) [2–4]. Inherent unprecedented GN features have resulted in escalated researches of other 2-D NMs for this application (transition metallic oxides/hydroxides, [5,6] transition metal dichalcogenides (TMDs), [7,8] hexagonal boron nitride (h-BN), [9,10] and transitional metallic carbides and nitrides (MXenes) [11,12]). Another group of NMs exhibiting similar features like GN and P—N are pnictogens (PTG), constituted of antimony (Sb), arsenic (As) BP, and bismuth (Bi) [13,14].

Relative to the timeline of P—N, in 2014, it was opined that P—N, as a novel elemental 2-D NM, displayed outstanding electrical features (mobility of $1000 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ with on/off current ratio of about 10^5 at

ambient temperature) along with anisotropic attributes having a tunable bandgap (0.31 eV–1.0 eV), thereby inculcating P—N with elevated prospects for use in electronic as well as optoelectronic gadgets. 2-D P—N bandgap lies between graphene (0 eV) and transition-metallic dichalcogenides (TMDs) (1.0 eV–2.0 eV) [4–6]. P—N photo-response at the infrared (IR) range, in combination with its elevated anisotropically light-matter interactivity, present P—N as novel research investigation in photonics, optoelectronics, and biomedically orientated based on 2-D NMs. In previous half decade, >1000 research articles linked BP/P—N have undergone publication, thereby revealing the elevated degree of enthusiasm geared towards BP/P—N-linked researches.

Nevertheless, GN has exhibited superior features in comparison with other 2-D NMs. On the other hand, the gapless defect of GN has limited its applications [13], thereby propagating further studies in exploring emerging 2-D NMs as key to these limitations, and prompting the advent of novel 2-D NMs with the recent emergence of phosphorene (P—N), exhibiting similar features with GN and providing panacea to the band gap deficiency [14]. Phosphorus (PN) exists in four prevailing allotropic forms (white, red, violet, and black) [15]. Fig. 1 presents the various allotropic forms of phosphorus [15].

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Black phosphorus (BP) is constituted of sheets connected by weak van der Waals forces (vdwf) capable of being used in formation of freestanding 2-D NMs, referred as phosphorene (P–N) [16]. P–N is similar to GN because they possess similar features as well as uses and have garnered great developmental strides attributed to their broad application segments, especially in energy exploration, and so [17].

Although P–N exhibits several outstanding features and application potentials, two key disadvantages inhibit its feasibility in usage. Firstly, is the inferior stability of 2-D P–N at ambient subjection. In spite of bulk P–N being very stable when compared with allotropes of phosphorus, the air prevalent oxygen and moisture reacts easily with P–N thereby destroying its crystal architecture on exfoliation to few-layered architecture [7]. Secondly, is the paucity of synthesis routes for broad-area elevated-quality few-layer P–N [8]. Presently, few-layered P–N is majorly fabricated via top-down processability of bulk BP crystals, difficult to use as a prevalent strategy of mass fabrication of few-layered P–N possessing ideal interfacial disposition.

The notable deficiency of few-layered P–N nanosheets is related to their inherent sensitivity towards oxygen as well as humidity in light presence, thereby making their usage at ambient conditions not possible. Varying strategies have been geared towards P–N sheets protection from oxidation. Nevertheless, P–N synthesis with improved stability to ambient parameters is still a very challenging segment of investigation. Another P–N drawback involves fabrication of single-layered P–N [9]. It is established that with a minimizing number of phosphorene sheets some of the physical features of materials went through some variations. Hence, in order to attain the most active NM possessing best catalytic as well as chemical features, it is beneficial to fabricate P–N nanosheets exhibiting a number of layers as minimal as feasible. However, numerous works has been conducted in this segment, yet garnering single-layered P–N remain challenging.

The parameters presenting 2-D NM as effective for EESDs includes its broad lateral dimension and hyper-thin features endowing it with ultra-elevated surface area as well as versatile presence of surfacial atoms, thereby offering efficiency for energy storage applications [19]. Secondly, its “completely surface” disposition provides the privilege of engineering features which are customized through surface modifications [20]. Thirdly, effective intercalation of ions along with energy storage in 2-D nanoparticulates within nanosheets via fast ion adsorption activities [21]. Fourthly, 2-D NM function as foundation for varying hybridized and hierarchical nanoarchitectures from 0-D towards 3-D, as no individual material can completely satisfy the stringent EESDs

requirements [22]. Moreover, their atomic thicknesses provide maximum mechanical flexiness and elevated packing density, hence facilitating their potential for the development of elevatedly flexible EESDs [23].

Furthermore, the inherently broad surface area, electrical conductivity and flexiness of 2-D P–N, present it as effective NM for EESDs. Nevertheless, the embedment of 2-D P–N within polymeric matrices forms P–N@polymeric nanoarchitectures. This improves the selective dispersion of nanoparticulates within a single phase out of the dual-phases or at the interfacial segment thereby enhancing the physicochemical features, passivation, air instability, thus, elevating its prospect as electrode substrate for electrochemical energy storage gadgets (EESDs) (supercapacitors (SCs), lithium ion batteries (LIBs) and emanating technologies (lithium–sulfur batteries (LSBs), magnesium ion batteries (MIBs), along with sodium ion batteries (SIBs)). Therefore, this paper, presents emerging advances in design, development, fabrication, characterization, electrochemical energy storage and conversion and photo-catalysts applications of phosphorene (P–N) and P–N polymeric nanoarchitectures (PPN). Currently, varying fabrication approaches have been utilized in preparation of P–N as summarily presented in Fig. 2 [18].

2. Strategies of P–N fabrication

2.1. Top-down approach

Phosphorene (P–N) fabrication approaches can be differentiated into top-down and bottom-up strategies. Top-down approaches are composed of BP (black phosphorous) separation into individual or a few arranged sheets, attained via mechanically affiliated or liquid affiliated exfoliation strategies. Moreover, these techniques are feasible due to layered architecture of the pristine BP bonded together by weak Van der Waals inter-sheet forces. Facile, inexpensive and reliable fabrication of P–N with superior features is paramount for future uses in a versatile range of segments. Notable achievements in P–N construction have been attained in previous two years. In subsequent segments, the benefits and deficiencies of present experimental strategies for synthesizing P–N and P–N-oriented materials are presented.

2.1.1. Mechanical exfoliation

Fundamental mechanically affiliated exfoliation entails the isolation of 2-D nanosheets utilizing an adhering tape. Individual sheets are

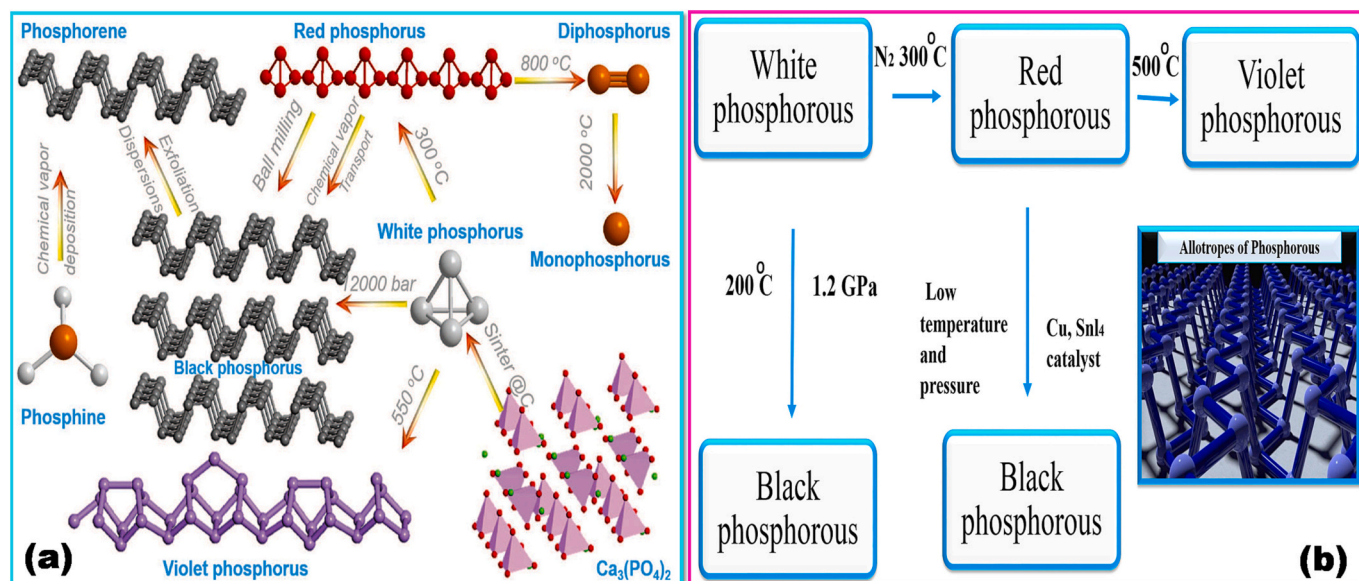


Fig. 1. (a, b) Allotropic forms of phosphorus [15].

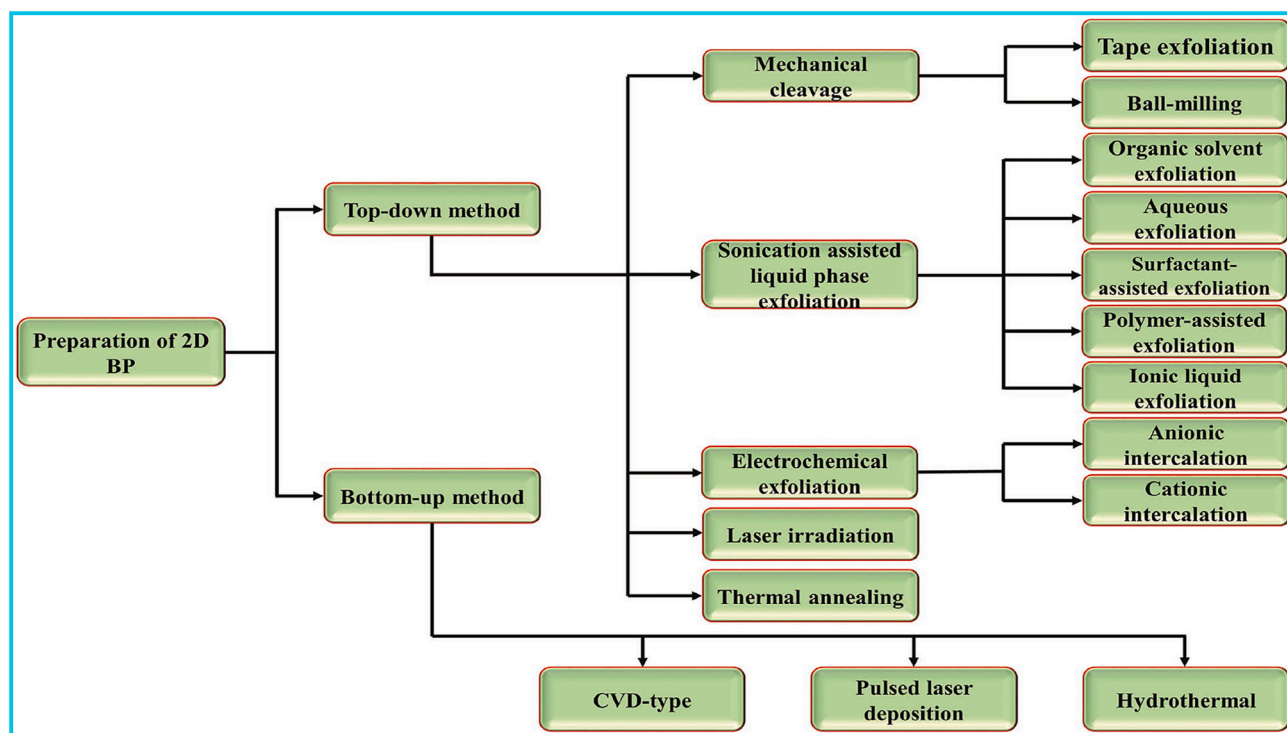


Fig. 2. Fabrication strategies of P–N [18].

conveyed upon Si/SiO₂ substrates, and subjected to cleaning utilizing isopropyl alcohol, methanol, and acetone, so as to eliminate residual scotch tape, with subsequent subsection to heat, so as to eliminate residual solvent [19]. The major deficiency of this fundamental technique include poor yield, hardline in garnering broad-sized single-sheet or few-layered BP (FLBP), attributed to BP instability in the presence of water and oxygen as well as the weak adhesive force upon SiO₂/Si substrates, in comparison with GN.

Presently, the fabrication of P–N is generally garnered through mechanical exfoliation. Nevertheless, this approach displays a versatile range of limitations (paucity of a geometrical control, poor thickness control, and dimension, inferior repeatability, and inferior production output, and critically garnerable via experimentation, thereby inhibiting its practical applications. Another notable challenge with the traditional mechanical cleaving approach is the left-over of chemical tracks on adhesive tapes post exfoliation. Thus, emerging mass fabrication techniques are essential for industrial uses.

2.1.2. Liquid-phase exfoliation

Liquid exfoliation is a potential strategy that has been versatily utilized in fabrication of ultra-flat 2-D nanosheets. This strategy has numerous techniques (oxidation, ion intercalation/exchange, or surfacial passivation via solvents) [10,11]. Hence, the end materials are fabricated in liquid dispersions composed of high levels of nanosheets as well as quantum dots. In addition, this strategy offers elevatable output of 2-D NM in solution with an effective, inexpensive, and simplified fabrication strategy. Hence, it has been effectively utilized in construction of varying 2-D NM, displaying outstanding performances in broad range of uses [12]. This strategy inculcates new horizons for mass fabrication of P–N nanosheets, capable of being vastly utilized in flexible electronics as well as optoelectronics gadgets.

2.1.3. Plasma etching

The gas plasma-enabled technology has been utilized in advancing mechanical exfoliation technique capable of producing single-sheets or few layers of P–N via heat ablation. A controlled micrometric resolution

argon plasma-facilitated approach, has been utilized in fabricating phosphorene mono-sheets via heat ablation [13]. Oxygen plasma dry etching for thinning thick exfoliated P–N sheets, layer by layer, using atomic precision, has demonstrated a novel elevatedly controllable strategy for constructing elevated quality, air-stabilized P–N films with assigned number of sheets ranging from a few to mono-sheets (Scheme 2). The synergy of aforementioned technique with mechanical cleavage gives an improved pathway for the regulation of P–N thickness, though not feasible for mass production [13].

2.2. P–N fabrication via bottom-up approach

2.2.1. Chemical vapor deposition (CVD) technique of P–N fabrication

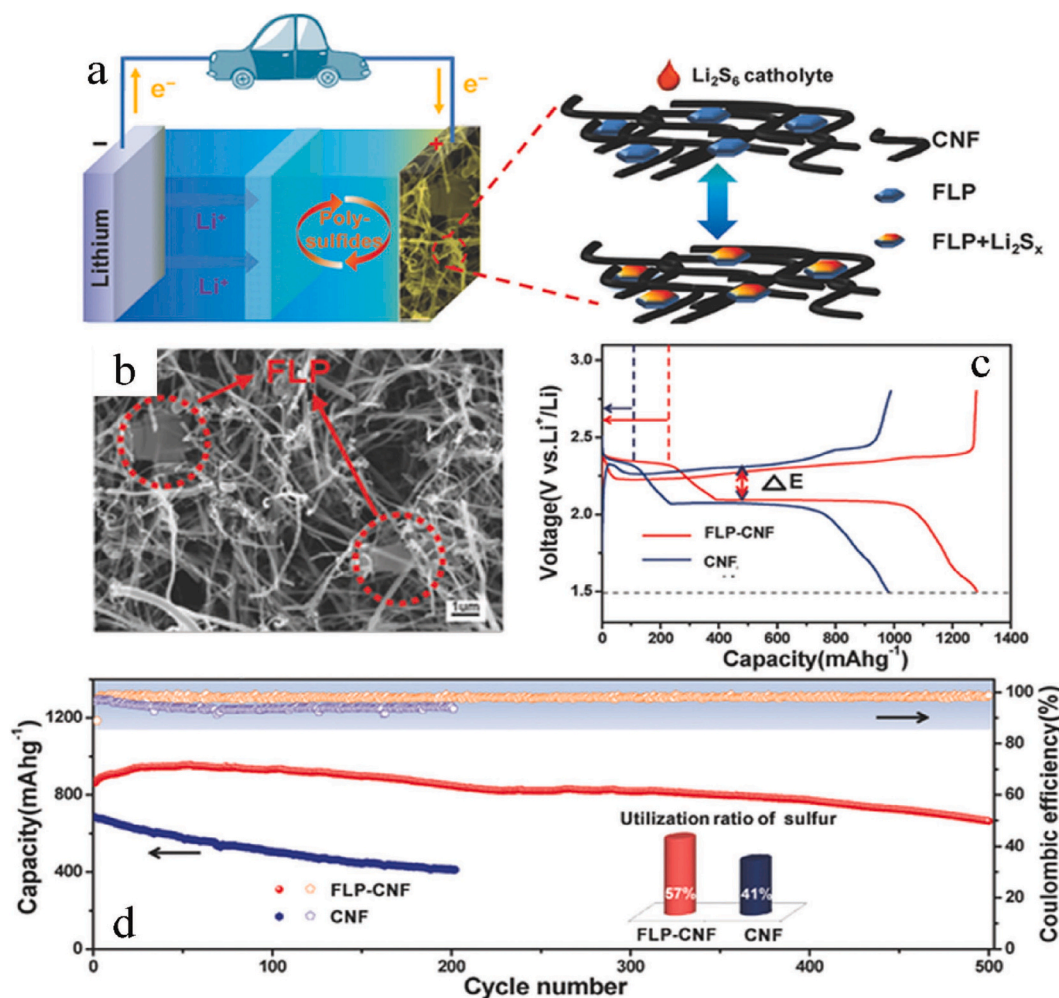
CVD is a unique bottom-up approach utilized in fabricating 2-D NMs on a specific substrate through precursors reaction at elevated temperatures. Here, this technique can result in a wide region ultra-flat 2-D NM with outstanding crystalline quality, tunability of thickness, as well as size. Presently, varying 2-D NMs, such as TMDs (MoS₂, WS₂, WSe₂), hBN, metallic oxides, and GN have undergone successful fabrication via CVD approach [14].

2.2.2. Mineralizer-facilitated gas-phase transformation technique

A study has reported the development of centimeter-scale selenium (Se)-inculcated P–N crystals with manipulatable doping composition utilizing a mineralizer-facilitated gas-phase transformation technique [100].

2.2.3. 2-D NM wet chemistry technique

Wet chemistry (hydrothermal, template synthesis, and self-alignment), are established strategies for 2-D material construction [101–104]. Recently, a study demonstrated a sublimation-enabled strategy for fabricating 2-D holey phosphorus-oriented nanosheets from bulky red phosphorous precursors via wet-chemical solvothermal reaction. Here, the mechanism of this technique entailed a solid–vapor–solid transformation propelled by a continual vaporization–condensation procedure, and subsequent bottom-up



Scheme 2. (a) Thick P—N sheet is initially exfoliated upon a SiO₂/Si substrate and the specimen modified using O₂ plasma etching (yellow spheres). (b) On O₂ plasma pre-treatment procedure, P—N sheets top layers undergo oxidation to PxOy, which operate as a protective barrier for the remnant P—N specimen underneath. (c) As a result of continual rate of etching, any specified number of P—N sheets down to a single-layer can be exactly prepared and the remaining sheet degradation is restrained due to the protective nature of the PxOy. (d) In order to enhance the shelf-life of the as-synthesized P—N, the specimen was also decorated with an Al₂O₃ protective sheet via ALD [13].

arrangement growth [105].

2.2.4. Stabilization of P—N

Few-layered P—N is not stable and undergo rapid degradation on subjection to ambient conditions. P—N utilization in varying applications is hindered by its rapid degradation on exposure to ambient environment. Varying groups have attempted to overcome this challenge [106].

2.2.4.1. P—N solvent exfoliation. Here, the degradation of P—N garnered from liquid-phase exfoliation (LPE) takes place more gradually when compared with that of P—N garnered via mechanical cleavage, which implies that P—N solvent exfoliation of BP is highly more effective technique for the fabrication of more stable P—N sheets. Solvent selectivity is imperative for garnering high quality P—N nanosheets [107].

2.2.4.2. Inorganic materials stabilization of P—N

2.2.4.2.1. P—N oxide stabilization. Recently, some passivation strategies have been utilized in stabilizing P—N sheets, such as decorating with Al₂O₃ via atomic layer deposition (ALD) [108,109]. In a study, oxygen plasma dry etching has been utilized in thinning down thickly exfoliated P—N sheets layer-by-layer with atomic exactness [110]. This

strategy represents a novel highly manipulatable strategy for developing high quality, air-stabilized P—N film with a specific number of sheets ranging from a few sheets to single-sheets. In a related study, ALD-garnered aluminum oxide (AlOx)-embedded P—N was devoid of any degradative-inducing bubbles post exposure to ambient environment while preserving the elevated carrier mobility and elevated on/off ratios of P—N FETs [111].

3. Electrochemical energy storage, energy conversion and photo-catalysts applications

The 2-D P—N inherent air stabilizing feature present it as a strong NM for a versatile range of uses. Additionally, 2-D P—N is an active electrochemical NM for effective energy storage mechanism (Fig. 3) [18], commencing from ions (Mg⁺, Li⁺, Na⁺) hosting to the surfacial accumulation of electrostatic charges in MIBs (as in electric double-layer capacitors (EDLCs)), as presented in Table 1 [24–32].

3.1. P—N application in LIBs

LIBs are renowned EESDs capable of reversibly converting chemical energy into electrical power and viable due to numerous benefits (elevated energy density, minimal self-discharge, with nil memory

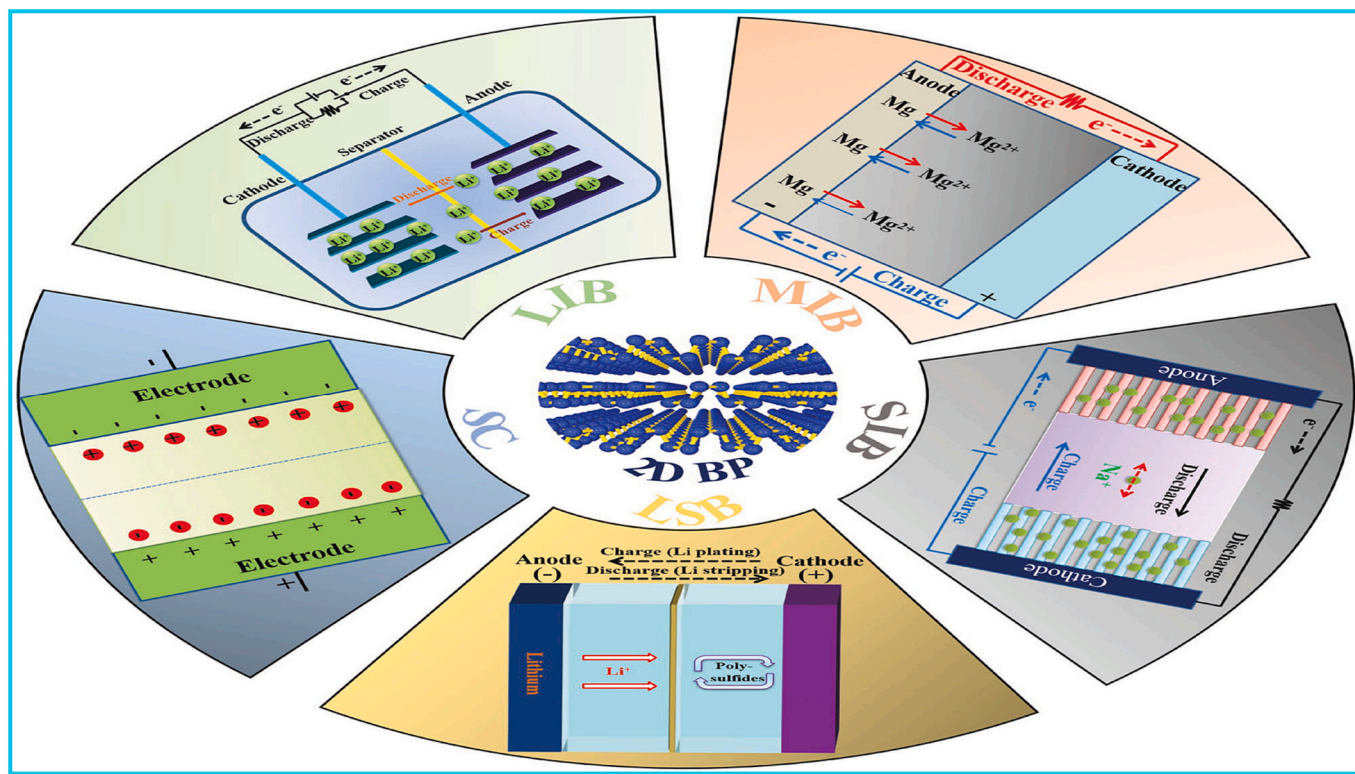


Fig. 3. Electrochemical energy storage of 2-D P—N [18].

influence), consisting of three parts, vis-a-viz electrolyte, cathode, and anode [33]. During operation, Li^+ ions continually move between a lithium-emitting cathode and a lithium-receiving anode [34]. When charging, Li ions undergo de-intercalation from the cathode, passing across the electrolyte, and intercalated within the anode. Here, the procedure undergoes reversion when discharging. LIBs performance critically depends on the quantity of ions housed per gram of the material referred to as the gravimetric capacity. In LIBs, the atomistic lithiation procedure in P—N is similar in comparison with graphite, where Li exhibit columnar intercalation situating in varying 2-D P—N sheets [35].

2-D P—N theoretically is an effective electrode for high-capacity LIBs because of the following effects: Firstly, Li atoms is capable of strongly binding with phosphorus atoms while maintaining a cationic status. Secondly, the diffusional energy inhibition of Li in the ZZ direction is very low (0.08 eV) [36], implying feasibility of ultra-rapid charging/discharging; Thirdly, 2-D P—N is capable of maintaining its architecture during lithiating and delithiating cycles, as the variation in volume is only 0.2 % [37]. Fourthly, a large average voltage (2.9 V) can be attained in 2-D P—N oriented LIBs and finally, the semi-conductive to metallic transition induced via Li intercalation of 2-D P—N resulting in effective electrical conductivity, thereby presenting 2-D P—N an effectual substrate for LIBs [38].

Furthermore, LIBs have garnered great interests because of inherently elevated storage capacity, satisfactory cycling features, and elevated energy density. Nevertheless, traditional graphite anodes as well as lithium metallic oxide cathodes situated on LIBs demonstrate theoretical specific energies of $\sim 400 \text{ W h kg}^{-1}$ and $\sim 200 \text{ W h kg}^{-1}$ (practical) respectively, which are not enough for satisfying the escalating quest for elevated-energy storage structures [39]. Hence, the fabrication of new inexpensive electrode substrates which are readily available has become paramount. Orthorhombic P—N has been proposed to be directly utilized in LIBs because of inherently layered architecture, which is similar to that of graphite. Moreover, P—N exhibit elevated bulk electrical conductivity (EC) ($\sim 102 \text{ S m}^{-1}$), miniature band

gap (0.34 eV), along with satisfactory density (2.69 g cm^{-3}), advantageous for attainment of both elevated power and energy densities [40].

In previous years, blackish and reddish P—N have garnered escalating attention due to inherent prospects as elevated-performance anode substrates in batteries. P—N undergo interaction with three Li atoms thereby forming Li_3P compounds, demonstrating elevatable theoretical specific capacity of 2596 mA h g^{-1} which correspond to Li_3P with outstanding cycling life of 100 cycles and 80 % capacity retention, which, seven times higher than graphite capacity (372 mA h g^{-1}) [41]. Putting in perspective, the elevated theoretical specific capacity and proper charge/discharge prospects (Table 2) [42–48], P—N is a potential anode material for LIBs.

P—N display a double sheet architecture along the (0k0) plane offering a 2-D interstitial space for LI insertion/extraction (Fig. 4a). The broadest space for LI intercalation occur along the a-axis in the (020) plane (Fig. 4b). The optimal space for LI insertion is along the (040) plane, along either the a-axis or c-axis, as displayed in the spaces projection among three close layers on the yz and xy plane (Fig. 4c). P—N undergo intercalation with three Li atoms in completely discharged state, hence exhibiting elevated specific capacity of 2596 mA h g^{-1} [49].

Hence, P—N mechanism of lithiation is depicted as follows: amorphous Li_xPy is garnered during charging, as crystalline Li_3P underwent stepwise transformation via Li_xPy during discharging. As the potential was minimized to 0.78, 0.63, and 0 V from the initial 2 V during lithiation, hence the multistep reaction is demonstrated as the conversion of $\text{P—N} \rightarrow \text{LiP} \rightarrow \text{Li}_2\text{P} \rightarrow \text{Li}_3\text{P}$, as presented in Fig. 5 [50].

3.2. Challenges of P—N application as LIB anode

Compared with silicon, fast capacity decaying ascribed to inferior EC along with severe volume variation during the lithiation/delithiating step is a critical challenge requiring panacea for P—N. Bulkish and micro-dimensioned P—N display low reversibility prone capacities, rapid capacity fading induced by volume elongation, and sluggish electron kinetics garnered from lithiation. Investigations have

Table 1

2-D P—N based NMs and P—N@polymeric nanoarchitectures for EESD [24–32].

NM	Fabrication strategy	Electrochemical behavior	Uses	Ref
2-D P—N/ PANI	Shear force milling	Capacitance of 354 F g ⁻¹ @ 0.3 A g ⁻¹ and 308 F g ⁻¹ @ 1.7 A g ⁻¹ were garnered, while a capacity retention of 96 % post 175 cycles @ 0.3 A g ⁻¹ were attained.	SC	[24]
2-D P—N/ PEDOT	Exfoliation in DMF	Capacity attained is 1397 mA h g ⁻¹ @ 100 mA g ⁻¹ and 370 mA h g ⁻¹ @ 10 A g ⁻¹ , while 67.4 % capacity retention post 100 cycles at 100 mA g ⁻¹ was garnered.	SIB	[25]
2-D P—N/ PEDOT	Exfoliation in DMF	Attained specific capacity is 1408 mA h g ⁻¹ @ 100 mA g ⁻¹ and 434 mA h g ⁻¹ @ 10 A g ⁻¹ , while capacity retention attained is 77.6 % post 100 cycles at 100 mA g ⁻¹ .	LIB	[25]
2-D P—N/ GN film	Exfoliation within deionized water	Capacitance of 37.5 F cm ⁻³ was attained @ 0.005 Vs ⁻¹ and 2.42 F cm ⁻³ at 1 V s ⁻¹ , with a maximum volumetric energy density of -11.6 mW h cm ⁻³	SC	[26]
2-D P—N film	Exfoliation in acetone	A capacitance of 17.78 F cm ⁻³ at 0.005 V s ⁻¹ and 1.43 F cm ⁻³ at 10 V s ⁻¹ , and a maximum volumetric energy density of 2.47 mW h cm ⁻³	SC	[27]
2-D P—N	Electrochemical exfoliation	Capacity of 1878.4 mA h g ⁻¹ at 100 mA g ⁻¹ was attained with 321 mA h g ⁻¹ at 2500 mA g ⁻¹ , while capacity retention of 102.1 % was attained post 100 cycles @ 1500 A g ⁻¹ .	SIB	[28]
2-D P—N/ GN	Exfoliation in NMP	Capacity of 2440 mA h g ⁻¹ @ 0.05 A g ⁻¹ and 645 mA h g ⁻¹ @ 26 Ag ⁻¹ was garnered, with 83 % capacity retention post 100 cycles @ 0.05 A g ⁻¹ .	SIB	[29]
2-D P—N/ GN hybrid paper	Exfoliation within deionized water	Specific capacity of 920 mA h g ⁻¹ @ 100 mA g ⁻¹ and 141 mA h g ⁻¹ @ 2500 mA g ⁻¹ were attained, while capacity retention is 80.2 % post 500 cycles @ 500 mA g ⁻¹ .	LIB	[30]
2-D P—N/ CF	Exfoliation in NMP	1262 mA h g ⁻¹ @ 0.2C and 785 mA h g ⁻¹ @ 3C (1C = 1675 mA g ⁻¹), as capacity retention is 73.5 % post 500 cycles @ 1C.	LSB	[31]
G/P—N/ G—G paper	Exfoliation in NMP	Attained specific capacity is 1633 and -656 mA h g ⁻¹ @ 100 mA g ⁻¹ and 1 A g ⁻¹ , respectively, while capacity retention is 85.8 % post 200 cycles @ 100 mA g ⁻¹	LIB	[32]

Key: P—N (Phosphorene); PANI (polyaniline); PEDOT (Poly(3,4-ethylenedioxythiophene)); GN (Graphene); CF (Carbon nanofiber); DMF (dimethylformamide); NMP (N-methyl-2-pyrrolidone).

propounded novel strategies for the enhancement of P—N EC which hinder cracking and pulverization [51]. Presently, traditional techniques include fabricating P—N nanocomposites through comingling with carbon black, GN, CNT, and other non-carbon NM, and utilizing fewer P—N sheets [52–54].

3.3. P—N/carbon nanocomposites for LIBs

In 2007, when P—N was initially used as anode, it exhibited very low coulombic effectiveness and fast capacity fading. Due to its poor EC (1 ×

107 S m⁻¹), garnered from its semiconducting attributes, it was propounded that P—N electrical performance as anode could be enhanced utilizing P—N@carbon nanocomposites. Hence, in a work, a set of xPyC (0 ≤ x ≤ 10, and x + y = 10) anodes underwent fabrication via ball-milling strategy for LIBs so as to ascertain the level of contribution of carbon in P—C nanocomposites [52]. 80 P—4C and 3P—7C specimens demonstrated outstanding initial reversible charge capacities (1803.5 and 1585.3 mA h g⁻¹), respectively, at 0.1 °C in voltage range (0.02–2 V), thereby showing that the overall capacity of xP—yC is function of not only for physically adsorbed P for elevated theoretical capacity, but also on the conductive carbon, resulting in a notable LI storage capability [52]. In another work, an approach for garnering strong phosphorous-carbon (P—C) bonds between P—N and GN was demonstrated [53]. P—C bonds (Fig. 6a and b) were garnered between P—N and carbon through a high energy mechanical milling (HEMM) technology under a 1.2 MPa Ar atmosphere [53].

4. P—N application in LSBs

Interests in LSBs have escalated due to elevated theoretical specific energy (2600 W h kg⁻¹) which is >5–7 times more elevated in comparison to traditional LIBs [42]. Elemental sulfur is plentifully available in nature, nil-toxic, and ecobenign, presenting LSBs viable for commercial usage [43]. The outstandingly elevated specific energy of LSBs is induced through synergy of two elements as active substrates, vis-a-viz metallic lithium anode (theoretical specific capacity: 3860 mA h g⁻¹) and elemental sulfur cathode (theoretical specific capacity: 1675 mA h g⁻¹) [44]. During discharging, the anodic reaction is the oxidation of lithium, Li → Li⁺ + e⁻, while the cathodic reaction is sulfur reduction, S + 2e⁻ → S²⁻ [45].

As a result of their high prevalence, inexpensiveness, poor redox potential (−2.71 V vs. standard hydrogen electrode), as well as similar route of chemistry in comparison with lithium, SIBs have garnered great attention like LIBs, especially when it comes to broad-scale energy storage [43]. With regards to SIBs, anode substrates should possess elevated sodium capacity, poor redox potential, elevated EC, elevated sodium diffusivity, as well as dependable structural stability.

In elucidation of the sodiation mechanism, P—N inferior cycling performance in SIBs makes it paramount for the exploration of the sodiation mechanism required for improving SIBs performance. Premised on the calculation (first-principles), solidation takes place through layered intercalation up to a composition of Na_{0.25}P, which induces P—N layers to slide [44].

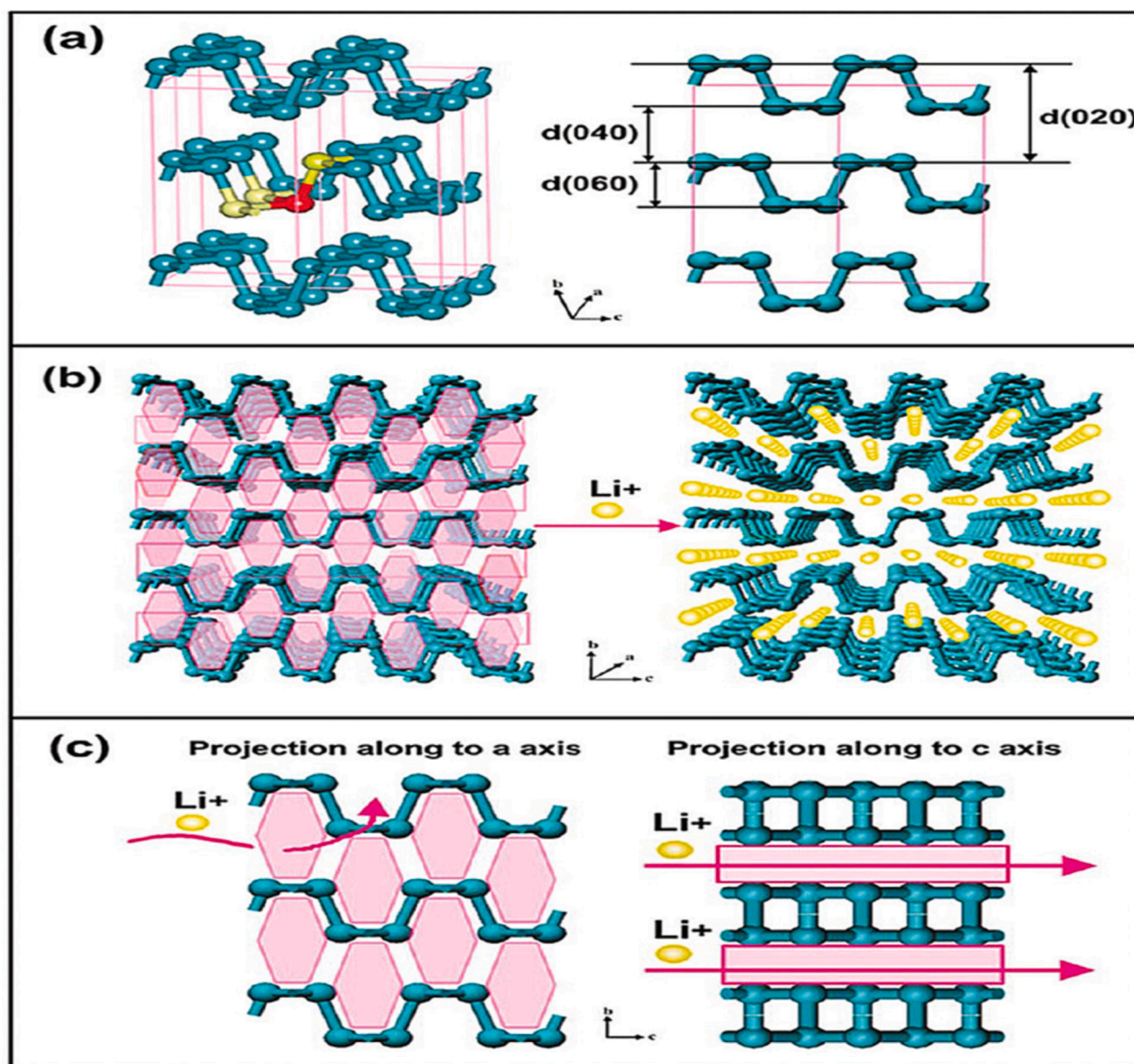
Though sheet-like architecture of P—N in SIBs offers improved kinetic (simpler Na atoms diffusion) as well as cycling performance (nil phase transition from layered to amorphous architecture), it exhibits the deficiency of electrolyte molecular entities ability for intercalating within layer spaces, resulting in layered architecture exfoliation. Therefore, P—N@carbon nanocomposites synergy are pathways for improving stability during sodiation.

Nevertheless, LSBs are impacted on by varying deficiencies including poor sulfur EC and its varying discharge entities, extended volume variation of sulfur on cycling, and the elevatedly soluble polysulfide intermediaries, which cause the “shuttle effect” in the course of electrochemical processes. A feasible panacea to these challenges involves sulfur encapsulation in appropriate matrix in order to increment the electrode conductivity, induce immobilization of the soluble polysulfide intermediates, while accommodating the sulfur volume variation. A study has initially evaluated the adsorption and diffusion disposition of varying polysulfide intermediates at differing lithiation/delithiating stations on single-layer P—N through theoretical computations, as another investigation depicted that the inclusion of 2-D P—N minimized the polarization, induced acceleration of the redox reaction, and improved the sulfur usage [42,43]. These results reveal that 2-D P—N could be used as appropriate and highly performing anchoring substrate for LSB cathodes. In a work, 2-D P—N was used in constructing an LSB in

Table 2

Electrochemical performances of P–N as anodes.

Electrodes	Electrolyte	Battery	Battery performance	Cycle performance	Ref
P–N/graphite NC with P–C bond	1 M LiPF ₆ in ethylene carbonate–ethyl methyl carbonate–dimethyl carbonate (EC–EMC–DMC) (1:1:1, volume ratio)	LIBs	Reversible capacity 2382 mAh g ^{−1} at 0.2C 1st coulombic efficiency 58 %	1849 mA h g ^{−1} after 100 cycles at 0.2C	[42]
P–N@GN	1 M NaPF ₆ in an ethylene carbonate (EC)–diethyl carbonate (DEC) solution with 10 % fluoroethylene carbonate (FEC)	SIBs	2440 mA h g ^{−1} at 50 mA g ^{−1} 2025 mA h g ^{−1} post 100 cycles at 50 mA g ^{−1}		[43]
P–N@GO paper with P–C bond	1 M LiPF ₆ in ethylene carbonate/diethyl carbonate (EC–DEC) (1: 1, volume ratio)	LIBs	1306.7 mA h g ^{−1} at 100 mA g ^{−1}		[44]
P–N@Ketjenblack-MWCNTs Nanocomposites	1 M NaPF ₆ in propylene carbonate (PC) with 2 vol% fluoroethylene carbonate (FEC)	SIBs		1700 mA h g ^{−1} after 100 cycles at 1.3 A g ^{−1}	[45]
Holy few layer P–N Nanosheets	1 M LiPF ₆ in ethylene carbonate/diethyl carbonate (EC–DEC) (1:1, volume ratio)	LIBs	1st cycle 2696 mA h g ^{−1} at 200 A g ^{−1}	1683 mA h g ^{−1} after 100 cycles	[46]
P–N/super P	1 M LiPF ₆ in ethylene carbonate/diethyl carbonate (EC–DEC) (1:1, volume ratio)	LIBs		600 mA h g ^{−1} after 1000 cycles at 100 mA g ^{−1}	[47]
Fewer layer P–N@GN	1 M LiPF ₆ in ethylene carbonate–ethyl methyl carbonate–dimethyl carbonate (EC–EMC–DMC) (1: 1: 1, volume ratio)		501 mA h g ^{−1} at 500 mA g ^{−1}	Capacity decay rate of 0.04 % per cycle at 500 mA g ^{−1}	[48]

**Fig. 4.** (a) P–N crystalline architecture. (b) Mobility pathways of Li ions within P–N. (c) Projections along the a-axis and c-axis of P–N electrochemical activity as anode material for LIBs [49].

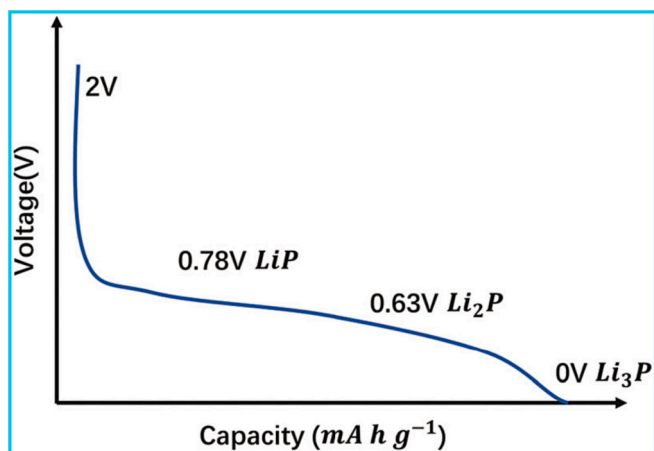


Fig. 5. Elucidation of lithiation multistep reaction [50].

synergy with polysulfides as presented in Scheme 1a. Here, FLP nano-sheets were embedded within a porous carbon nanofiber (CNF) architecture (Scheme 1b) via comingling of CNFs with liquid induced exfoliated 2-D P–N subsequently by vacuum filtration [18].

4.1. P–N@carbon nanocomposites for LSBs

A P–N/Ketjenblack-MWCNT nanocomposite fabricated via HEBM exhibited elevated initial coulombic efficiency (>90 %), elevated specific capacity, and outstanding cyclability at an elevated rate of charge/discharge ($\sim 1700 \text{ mA h g}^{-1}$ post 100 cycles at a current density of 1.3 A g^{-1} situated on the mass of P with elevated phosphorus composition (70 %) [45].

Furthermore, a study on electrode/electrolyte P–N interface electrode using polyacrylate binder has been conducted to decipher creation of a solid-electrolyte interphase (SEI) in alkyl carbonate ester and its emergence during cycling [46].

5. P–N application in MIBs

A viable route of utilizing post-lithium ion entails replacement of lithium with magnesium (Mg). Here, metallic Mg is effective anode material for rechargeable batteries because of inherent beneficial features (inexpensiveness, ease of availability, elevated melting point (649°C), a minimal potential of -2.37 V , as well as elevated volumetric capacity of $3833 \text{ mA h cm}^{-3}$ [46]. The theoretical specific capacity of Mg is 2205 mA h g^{-1} , offering two electrons per atom. In a reported MIB system, Mg ions were deposited/stripped upon the anodic side and

intercalated/de-intercalated on the cathodic side [47].

Varying challenges have hindered MIBs commercialization especially electrolytes. In spite of these challenges, MIBs are still attractive substrates for EESDs. In a study, P–N was shown to be utilized as an electrode substrate for MIBs as it was capable of storing Mg atoms up to Mg_2P . Prevalence and elevated safety features present MIBs as attractive substrate for meeting energy requirements of broad-scale gadget uses. Nevertheless, paucity of compatibility between anode–electrolyte, miniature electrochemical platform of electrolyte solutions, gradual Mg^{2+} cations solid-state diffusion, as well as the relatively poor specific energy of Mg batteries remain challenges [48].

6. P–N application in SIBs

SIBs depicting as possible replacement for LIBs, have garnered notable attention due to the prevalence and low cost of sodium entities [49]. For SIBs, copper can be replaced by aluminum as anodic current garnerer as sodium cannot be subjected to alloying interaction with aluminum at low voltage, thereby presenting SIBs as affordable [50]. Here, the basic foundation of SIBs is similar to those of LIBs, whereby, sodium ions are moved within two electrodes (cathode and anode) via an aqueous/non-aqueous sodium-ion electrolyte housed within the two electrodes on charging and discharging [51]. On intercalation of the sodium atom within bulkish P–N, a relative glide of $1/2(010)$ took place, causing atomic stacks reordering from AB to AC (Fig. 7) in P–N [52].

Presently, theoretical investigations have revealed that 2-D P–N is a potential anode substrate for SIBs [52]. P–N display greatly layered architecture compared to that of graphite, implying that sodium is capable of being stored between the 2-D P–N sheets [52]. Inherently strong binding energy within sodium–2-D P–N suggest that sodium can undergo stabilization upon 2-D P–N surface without clustering.

6.1. Electrochemical reaction mechanism of sodium ion batteries

The P–N inferior cycling behavior in SIBs induces their imperativeness for exploration of the sodiation mechanism to enhance their performance in SIBs. Relying on first-principles evaluation, sodiation takes place through the sheet-like intercalation to about a constitution of $\text{Na}_{0.25}\text{P}$, which induces sliding in P–N sheets [49,50]. A single Na atom tends towards binding four P atoms with appropriate energy of about 0.08 eV per atom, enhancing easiness for the sliding of P–N layers. On inclusion of two more Na atoms to $\text{Na}_{0.25}\text{P}$, the cleavage of P–P bond commences. Thus, finally Na_3P , only low co-ordinate P entities are recovered. Summarily, when Na concentrations are beyond a certain constitution of $\text{Na}_{0.25}\text{P}$, P–N sodiation mechanism varies from intercalation within P–N sheets to synergizing with P atoms (Scheme 3) [53].

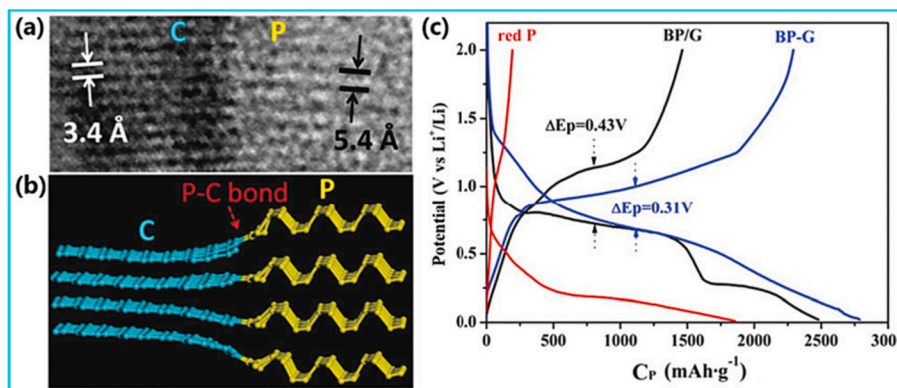
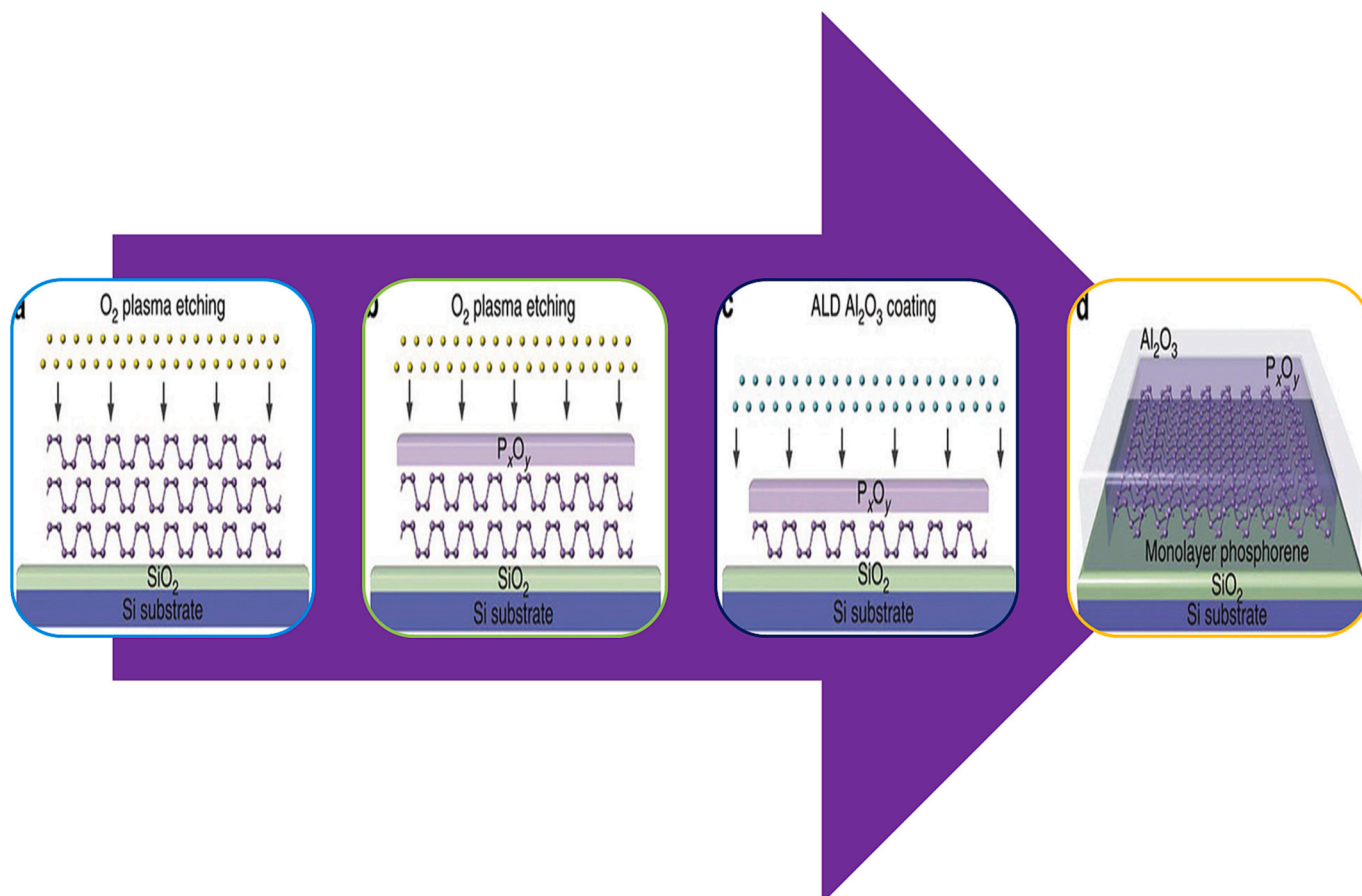


Fig. 6. a) C–P bond TEM image (b) Schematic elucidation of C–P bond. (c) Voltage outlay of reddish P, P–N@G, and P–N@G electrodes for initial cycle between 0.01 and 2 V at a current density of 0.2 C [53].



Scheme 1. a) Elucidation of FLP@CNT matrix utilized as host for lithium polysulfide catholyte. b) SEM of FLP@CNT. (c) Galvanostatic charge–discharge voltage outlay of initial cycle at 0.2C. d) Cycling stability and CE. The inset depicts sulfur usage at a current density of 1C (calculated based on the maximum capacity during cycling; the theoretical capacity is 1675 mA h g⁻¹) [18].

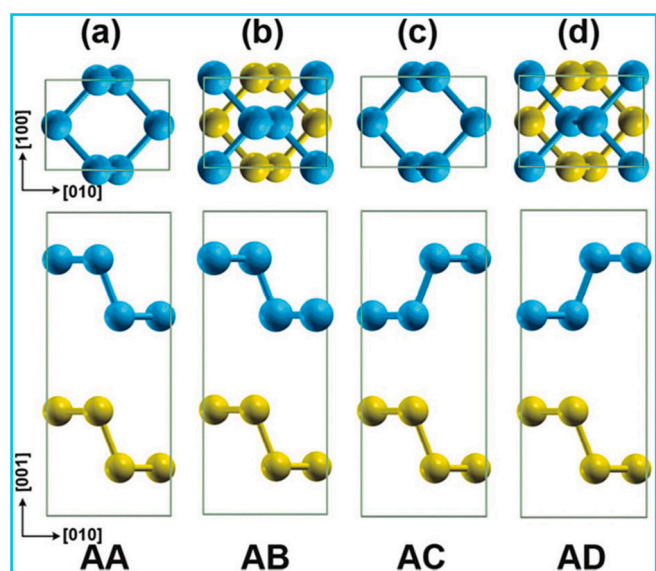


Fig. 7. P–N Top and side views for (a) AA, (b) AB, (c) AC, and (d) AD stacking arrangement [52].

The order of stacking has been demonstrated to play a key function in electronic attributes of layered entities, like graphite and MoS₂. P–N layered architecture in SIBs offers enhanced kinetic (better diffusion of Na atoms) as well as cycling behavior (no phase transition from layered

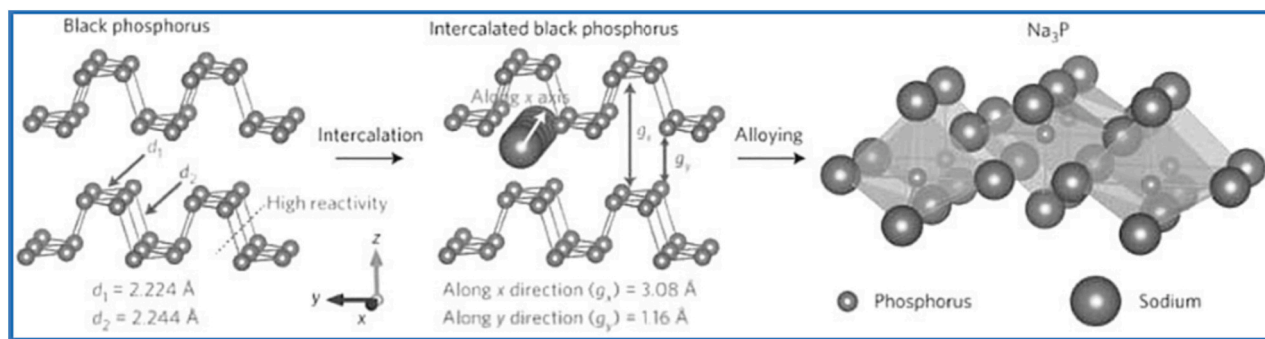
to amorphous architecture), it persists in displaying the negative effect of electrolyte molecular entities ability to induce intercalation within layer spaces, resulting in the exfoliation layered architecture. P–N–carbon nanocomposites are an alternative for improvement of stability during sodiation [54].

7. P–N application for SCs

SCs or electrochemical capacitors or ultra-capacitors have garnered great research attention recently attributed to their elevated power density, extended cyclic stability, as well as hyper-rapid charging–discharging rate. Placing in perspective various charge storage mechanisms, they can under classification into two forms, vis-a-viz (i) EDLCs and (ii) pseudocapacitors [55]. For EDLCs, energy undergo storage via electrostatic charge gathering at the interface of electrode–electrolyte thereby resulting in a double layer. Here, charges are dispersed via physical absorption within EDLCs. Hence, they are unlimited by electrochemical charge transfer kinetics of batteries operating at charge–discharge rates in level of seconds [56].

Carbon nanomaterials (for instance, activated carbon) prevalently utilized as electrodes for EDLCs because of outstanding features, relative to poor electrical resistance, elevated surface area, chemical inertness, along with stability. Hence, Pseudocapacitors store energy via rapid and reversibility of surfacial redox interactions between electrodes active material and the electrolyte [57].

Metallic oxides and conducting polymeric matrices are notable pseudocapacitive materials. Such electroactive entities facilitate elevated energy densities relative to EDLCs. Nevertheless, elevated energy density is garnered in place of low rate capability and minimized



Scheme 3. P–N elucidation prior sodiation, with initial step of sodium-ion intercalation, and subsequent step of synergizing to form Na₃P [53].

cycle life. Surficial area as well as intrinsic conductivity are two electrode parameters of an electrode material utilizable for EDLCs [58].

The intrinsic features of 2-D P–N position is a critical material for SC applications. Through distribution of exfoliated P–N nano-sheets in acetone, it was shown that a facile drop-casting technique to construct a flexible P–N films [59]. Hence, P–N films exhibit unique electrochemical double-layer capacitance feature. Thus, it was shown that flexible micro-SCs relying on the inter-digitalized hybrid electrode pattern of 2-D P–N nanosheets and GN (PG-micro-supercapacitors (MSCs) [60]. PG-MSCs exhibited a maximum volumetric energy density of $11.6 \text{ mW h cm}^{-3}$ due to synergism between 2-D P–N and G-N [60].

In another investigation, 2-D P–N/polyaniline (PANI) hybrid nanocomposites for pseudocapacitors was constructed [61]. Hence, the specific capacitance of 2-D P–N@PANI hybrid nanocomposites attained 354 F g^{-1} at a current density of 0.3 A g^{-1} in comparison with 308 F g^{-1} of PANI material. The improved performance is ascribed to the broad surface area of 2-D P–N nano-sheets, offering a broad surface area for the nucleation of PANI and enhanced accessibility of ions to PANI active material [62].

Although 2-D P–N displayed notable prospects as an electrode substrate in SCs, whereby its capacitance has been improved for versatile range of challenging uses. Pore architecture, such as SSA along with pore size dispersion, is paramount parameter which notably affect the performance of electrode materials. The construction of porous 2-D P–N can be an effectual strategy to enhance performance. Furthermore, doping is an effectual strategy to improve the pseudocapacitance of P–N. Broad SSA, as a 2-D NM, is regularly lost because of restacking, as well as exceptional conductivity which are compromised due to absorbed impurities as well as defects [63].

Hence, in a 3-D structure, the restacking of 2-D P–N are broadly eliminated, and effective electrolyte diffusion can be attained. Nevertheless, fast and continual path-routes for electronic mobility can be offered due to close contact between 2-D P–N sheets [64]. Hence, the construction of 3-D P–N electrode architecture is another approach for enhanced performance. Interaction investigation between 2-D P–N and other 2-D NMs has opened up a new horizon for the development of advanced 2-D P–N-based electrodes [65].

SCs have garnered great attention as a result of their outstanding performances (elevated power density, fast charge/discharge rate and long cycle life [66]. With evolution of P–N dispersions [67] using drop coating strategies, [68], P–N films have been fabricated using PET substrates (Fig. 8) and a SC depicting a capacitance of 13.75 F cm^{-1} at a scanning rate of 0.01 V s^{-1} was garnered [68].

These dual electrodes of P–N were demonstrated flexibility and minimal loss of current density. Here, maximum current density as well as power density attained were $2.47 \text{ mW h cm}^{-3}$ and 8.83 W cm^{-3} respectively. In future, it is anticipated that optimized fabrication strategies will enhance P–N highly interesting for energy gadgets in wearable electronics.

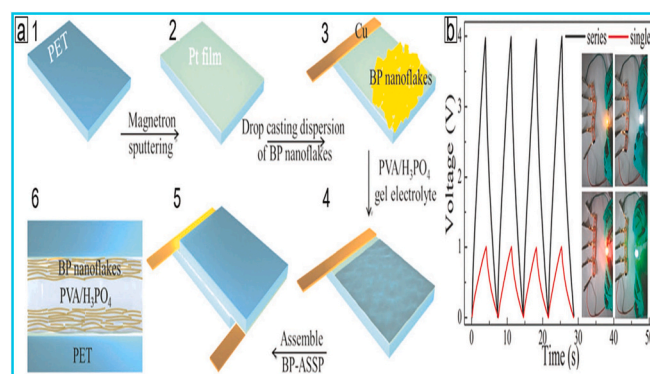


Fig. 8. Black phosphorus as electrodes in flexible supercapacitors. a) Schematic of the fabrication of all-solid-state supercapacitors. b) Galvanostatic charging/discharging profiles for single and 4-series supercapacitors [68].

8. P–N application in photovoltaic gadgets and solar cells

In a traditional solar cell, the semi-conductive p–n intersection differentiates the photo-garnered electron–hole pairs, whereas the dual terminal electrodes gather charges and take them to exterior circuits (integrated grids and storage batteries). These photovoltaic gadgets exhibit remarkable photoelectron effectiveness and extended stability. Nevertheless, their flaws emanate from either toxicity cadmium metal or limited presence of indium metal, in addition to intricate manufacturing ethics. [69,70]. These challenges have been mitigated by emerging photovoltaics (organic solar cells, dye-sensitized solar cells, perovskite solar cells, and van der Waals hetero-architectural solar cells) [71–73].

Recently, non-vacuum construction strategies were utilized in depositing adequate materials for optical absorbers whereby spin coating was used in depositing conjugated PEDOT: PSS (electron donor) in an organic solar cell, and nano-crystalline Cu (Zn, Sn) S₂ film for a CZTS solar gadget [74]. Hence, P–N has depicted as ideal material for photovoltaic application with the benefit of elevated EC [75] and easy tunable band gaps [76].

P–N quantum dots have been utilized in bridging PEDOT: PSS and perovskite layers for potentially enhanced performance in organic solar cells, dye-sensitized solar cells, and hetero-architectural solar gadgets [77–79]. Moreover, the external quantum effectiveness as well as current density-voltage curves can be enhanced significantly (Fig. 9) [80]. Due to inherent heat sublimation, inclusion of P–N use in device construction can be carried out at low temperature procedures ($<400 \text{ }^{\circ}\text{C}$) [80]. Generally, P–N embedment is capable of enhancing gadgets performance efficiency in various types of solar cells, as implied by varying theoretical reports [77–80], though paucity of experimental works abides.

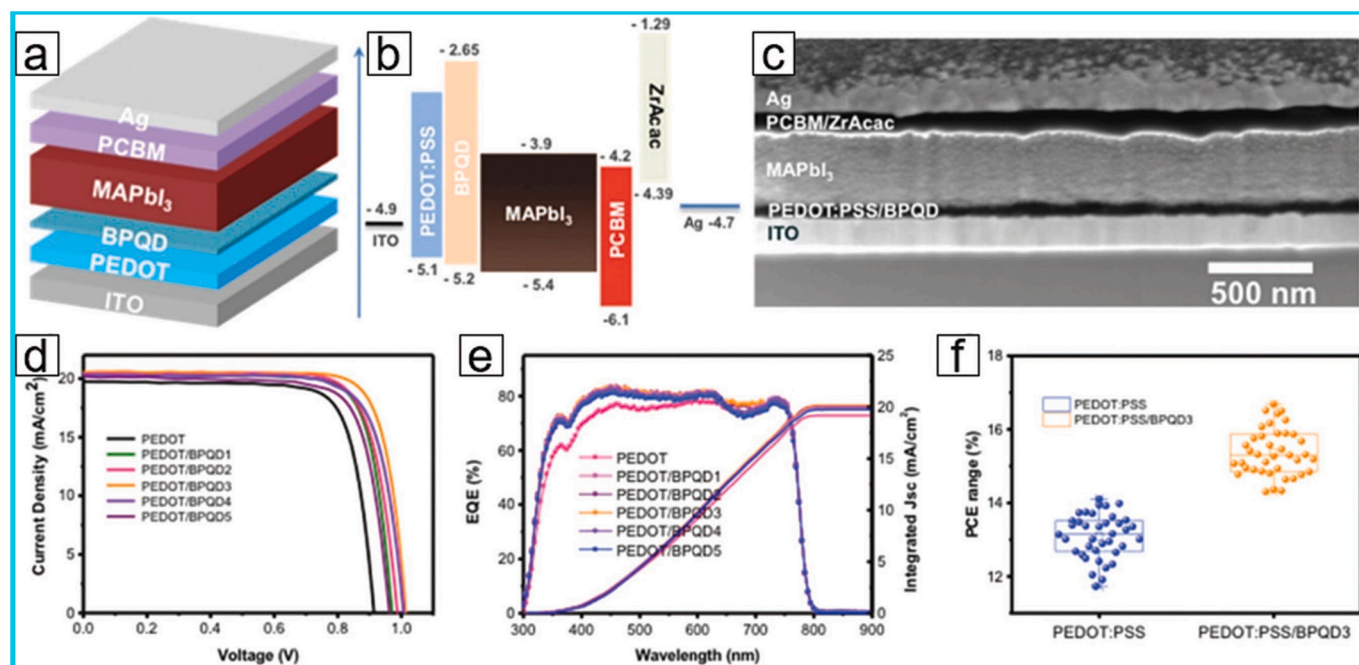


Fig. 9. P–N for perovskite solar cells. a) Gadget configuration of planar perovskite. b) Schematics of the gadget energy level diagram. c) SEM image of the c.s. of solar cell. d) Current density–voltage features of P–N quantum dot focused on perovskite solar cell. e) External quantum efficiency. f) Photoelectron conversion efficiency of solar cells with and without P–N quantum dots [80].

9. P–N photo-catalysts for solar fuel fabrication

Photo-catalysts are gadgets capable of converting light energy into chemical energy, either minimizing or oxidizing, via a redox reaction [81]. The photo-propelled chemical reactions proceed through transferring of charge conveyors, where solar light garners electron-hole pairs in a semi-conductor surface. A schematic elucidation of redox

potentials for H_2 production and CO_2 reduction is presented in Fig. 10 [82].

Photocatalytic electrochemical reaction critical steps include light absorption along with charge carrier excitation [83]. Here, light use effectiveness evaluating the solar energy conversion, depends on the band architecture of the photocatalytic semi-conducting devices. Four typical pathways for photocatalytic solar energy conversion include;

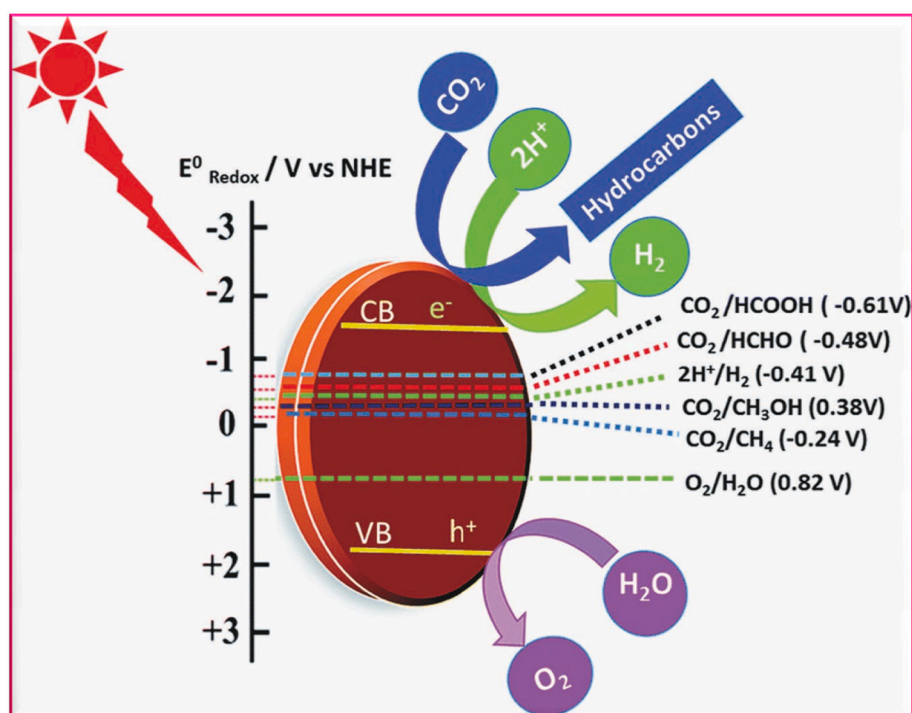


Fig. 10. Schematic energy diagram of photocatalytic semiconductor for water splitting and carbon dioxide reduction reactions. The redox potentials are referenced with normal hydrogen electrode (NHE) at pH = 7 [82].

Firstly, water splitting into H_2 and O_2 representing a typical pathway for photocatalytic solar energy conversion (Fig. 11) [84].

Secondly, photo-catalyst for CO_2 reduction converting light energy into solar fuels releasing combustible gases (CH_3OH , CH_4 and CO) [85–87]. Thirdly, photocatalytic hydrogenation is capable of raising the heating level of fuels for combustion, thereby maximizing the fuel utilization effectiveness. In an investigation, DFT computations were utilized in proposing that 2-D P–N and boron dopants as P/B Lewis pair functioned as an effective catalyst for ketones, nitriles and ethylene hydrogenation [88]. B-doped P–N enabled a dual-step hydrogenation reaction (Fig. 12) [88].

10. P–N electro-catalyst for oxygen evolution reaction

The oxygen evolution reaction (OER) [89,90] and oxygen reduction reaction (ORR) [91,92, 93] are essential in electrochemical energy conversion as well as storage, especially for fuel cells [94,95] and metallic-air batteries [96,97]. OER is a key procedure with respect to electrocatalysts used in water splitting gadgets. In both reactions, the electro-catalyst must exhibit good electrical charge conductivity, wide specific area, as well as elevated activity.

11. P–N for thermoelectric converting applications

A thermoelectric converting gadget is a solid-state gadget capable of converting heat-flux to electricity using the Seebeck effect [98]. For thermoelectric effect, additionally referred as the Seebeck effect, a temperature gradient within a heat conducting material results in heat flow, propelling charge transport diffusion [99]. Here, P–N nanoribbons displayed enhancement of the Seebeck coefficient (Fig. 13) [99].

The anisotropic architecture of P–N depicts elevated Seebeck coefficient, elevated EC, elevated power factor, and reduced thermal conductance in comparison with zigzag directions suggesting that P–N

nanoribbons are potentially elevated performance materials for thermoelectrically affiliated applications. Fig. 14 presents potential energy storage and conversion applications of P–N. Meanwhile, a number of researches have fabricated a range of polymeric nanoarchitectures for multifacet applications [112–184].

12. P–N for potassium-ion battery (KIB)

The scarce availability and paucity of lithium has enjoined high quest for developing novel rechargeable batteries utilizing alternative charge conveyors. Potassium-ion batteries (KIBs) are forms of batteries which are analogous to LIBs, and use potassium ions during charge transfer in place of lithium ions [185]. KIBs are topping the list of available alternatives due to the abundance of their raw materials, relatively elevated energy density, rapid ion transport kinetics within electrolyte, and relative inexpensiveness [186].

Nevertheless, numerous challenges inhibit KIBs development, such as inferior reversible capacity, low rate performance, and poor cycle stability and these have hindered their versatility of application [187–190]. Firstly, the broad potassium ion presents the volumetric variation during the charge/discharge procedure more notable for PIBs in comparison with other alkali metallic-ions batteries, which induces architectural instability within the electrodes. Secondly, K^+ low diffusivity in solid electrode materials inhibits rate performances [191]. Finally, KIBs electrolyte suffers from severe degradation with numerous parasitic reactions ascribed to elevated potential of K^+/K redox [192]. Therefore, the quest for KIBs development has escalated [193].

12.1. Anodes for KIBs

Similar to LIB, graphite is capable of accommodating potassium intercalation within electrochemical procedures [194, 195]. Other nanocarbon materials besides graphite has also been utilized as anode substrate for KIBs (expanded graphite, carbon nanotubes, carbon

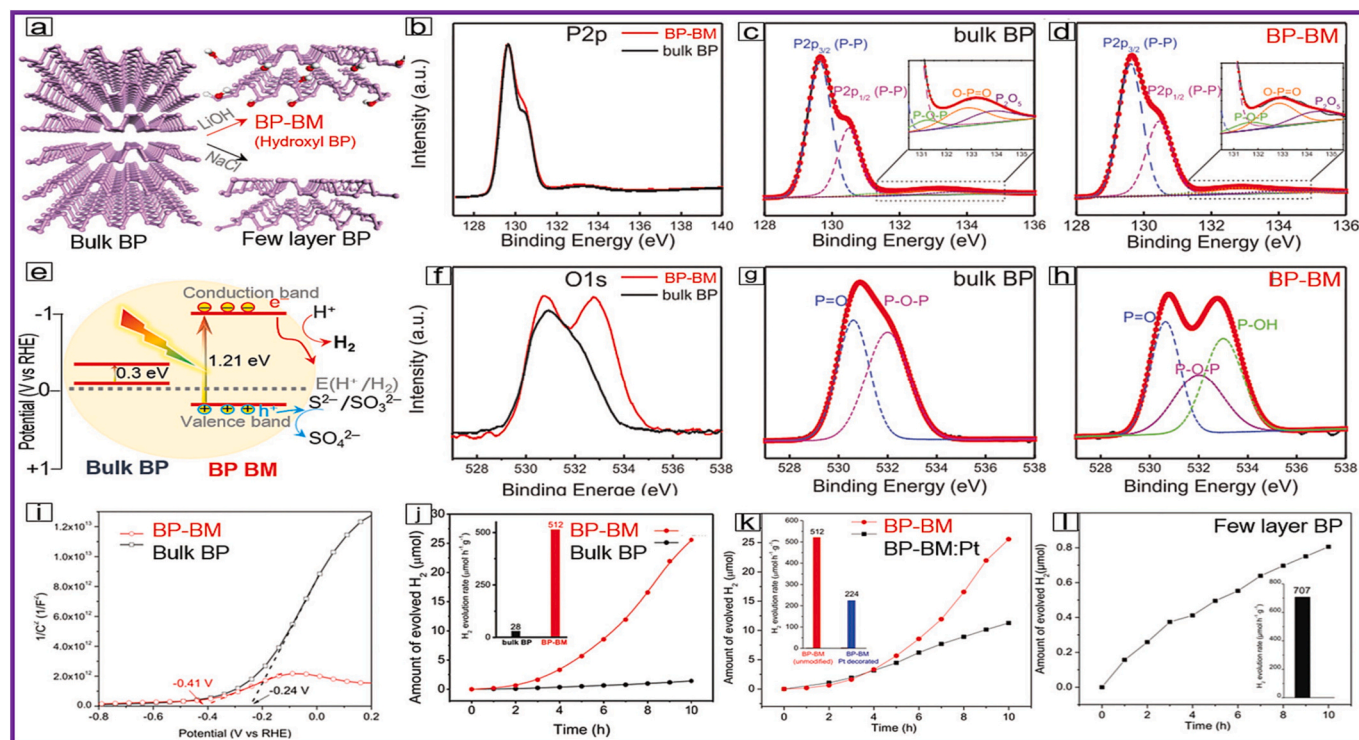


Fig. 11. H_2 generating efficiency of P–N photo-catalyst for water splitting. a) Illustration of functionalization of P–N. (b, c, d) XPS spectra of P 2p of bulkish P–H and OH functionalized P–N. e) Schematic elucidation of photocatalytic H_2 evolution. f, g, h) XPS spectra of O 1s for varying P–N forms. i) Mott–Schottky plots of black P–H and functionalized P–N at a frequency of 1 kHz. j,k,l) H_2 production outputs of varying forms of P–N [84].

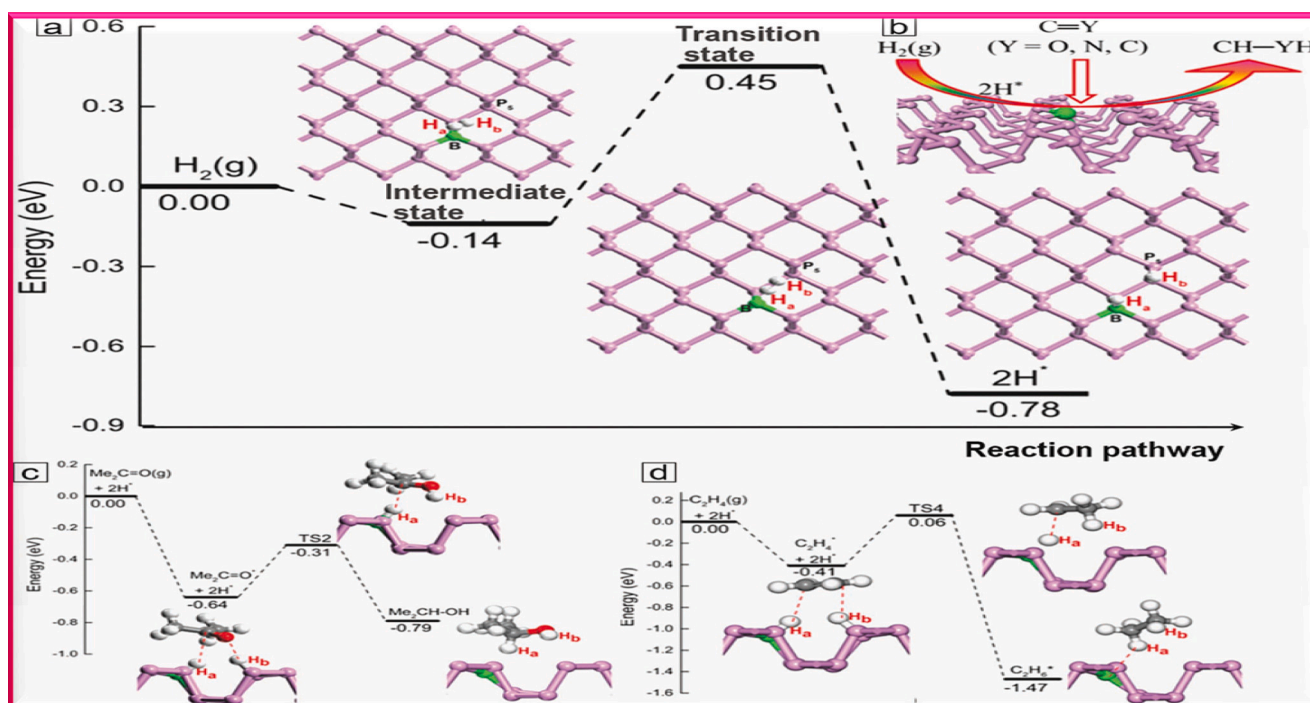


Fig. 12. P—N surface hydrogenation. a) H_2 dissociation route of reaction for P—N. b) Schematic elucidation of P—N catalyzed alcohol hydrogenation and hydrocarbon manufacturing from unsaturated micro-molecular entities. c) Ethanol ($\text{C}_2\text{H}_5\text{OH}$) with d) ethane (C_2H_6) manufacture reaction route [88]. Finally, photocatalytic detoxification of wastewater is capable of energy consumption minimization of water recycling treatments (photocatalytic water disinfection) [89].

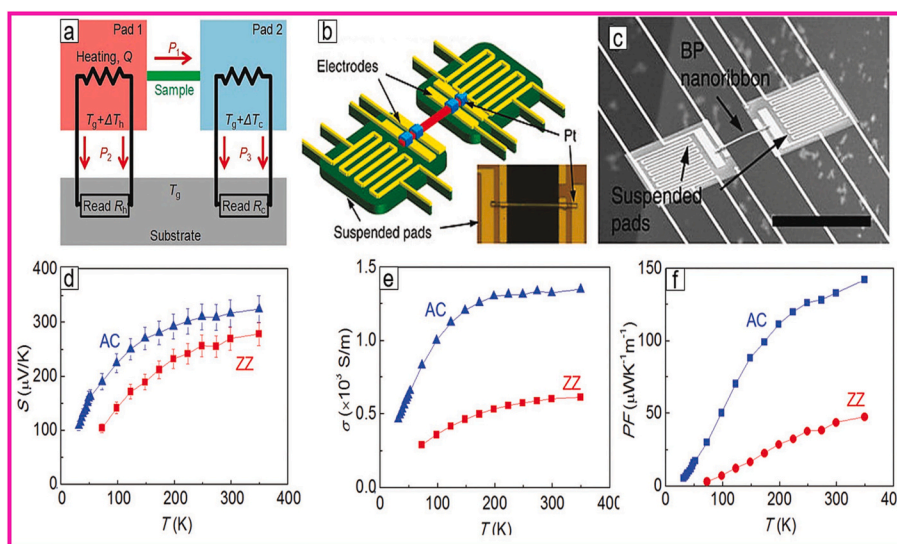


Fig. 13. P—N nanoribbons thermoelectric performance experimental evaluation. a) Schematic depiction of thermal mobility measurement. b) Elucidation and optical graph of P—N nanoribbon in a suspended-pad micro-gadget. c) thermoelectric gadget SEM image. d, e, f) Seebeck coefficients, electrical conductivity, and power factor, for armchair and zigzag ribbons with temperature dependence [99].

nanofibers in addition to nitrogen or phosphorus-decorated carbon materials) [196].

12.2. Cathodes for KIBs

Asides conventional Prussian blue cathode along with its analogs, studies on cathode component of KIB dwell on solid ionics and nano-architectures. A number of potassium transition metallic oxide have been shown as cathode material with a layered architecture [197] Poly-anionic entities possessing inductive defects could offer the highest

working voltage among other forms of cathode for KIBs. During the electrochemical cycling procedure, its crystal architecture will undergo distortion forming more inculcated defects on insertion of potassium ion [198].

12.3. Electrolytes used for KIBs

As a result of chemical activities more elevated in comparison with lithium, electrolytes used for KIBs requires careful engineering so as to mitigate safety issues. Ionic liquid electrolytes provide a novel route of

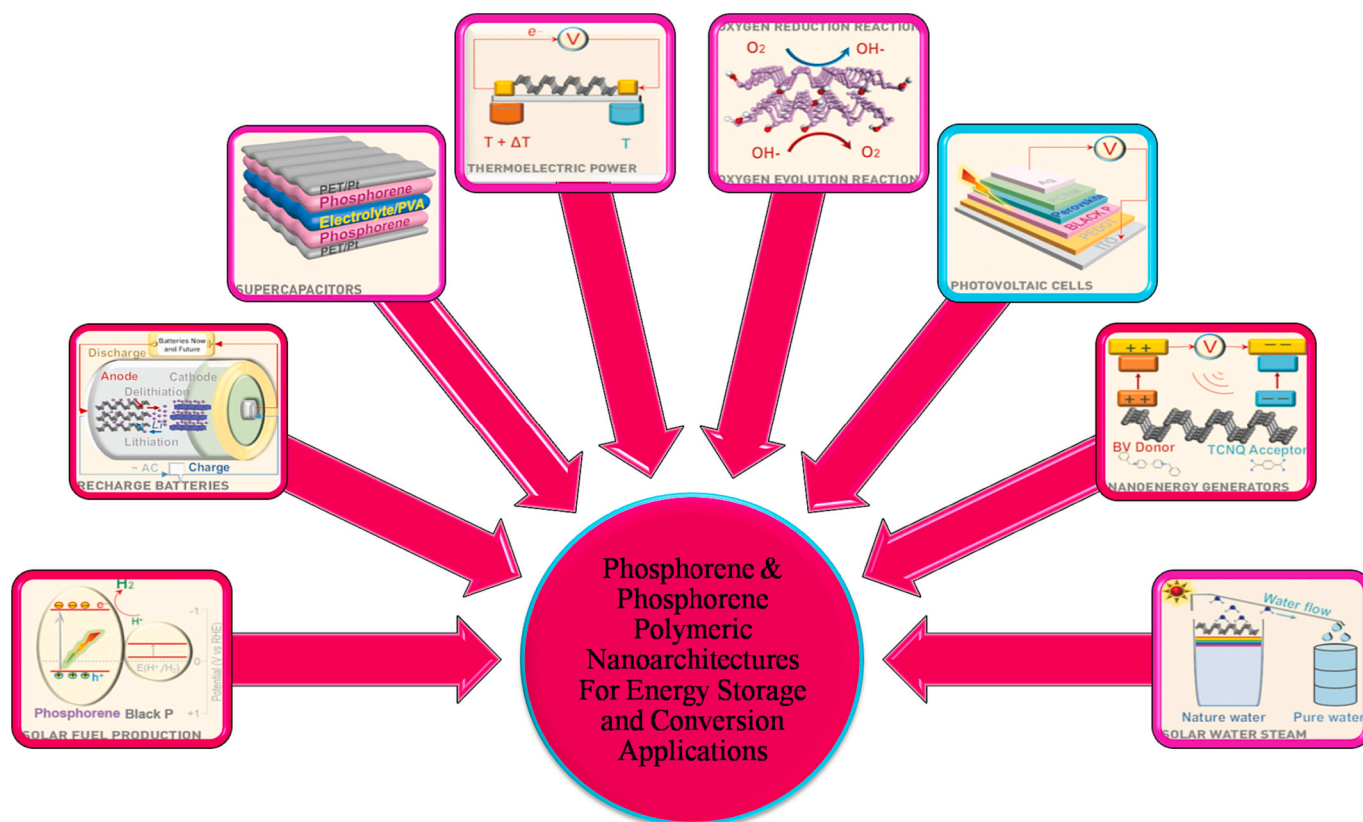


Fig. 14. Potential energy storage and conversion applications of P–N and P–N polymeric nanoarchitectures.

enlarging electrochemical activities of potassium ion battery with higher negative redox voltage and it's specifically stabilized using graphite anode [199]. Recently, polymeric electrolytes for all-solid-state KIB have garnered great attention attributed to inherent flexibility and improved safety [200].

12.4. Advantages of KIBs

In addition to Na-ion, K-ion is the key chemistry alternative substrate for LIB [201]. The KIBs exhibit certain benefits when compared with Li-ion (e.g., LIBs), where the cell designing is simplified and both the material as well as the fabrication steps are inexpensive. The major benefit is the prevalence and inexpensiveness of K in comparison with Li, thereby making KIBs a prospective material for large-scale batteries like household energy storage as well as electric vehicles [202]. A notable advantage of KIBs over LIBs is potentiality for a more rapid charging [203].

12.5. Applications of KIBs

In 2005, a KIB utilizing molten electrolyte of KPF6 was patented [204,205]. A Chinese company Starsway Electronics marketed the first KIB-powered miniature media player as a high-energy gadget [206]. KIBs have been presented for large-scale energy storage attributable to its outstanding cyclability, but present prototypes only withstand a hundred charging cycles [206,207]. As at 2019, five major challenges were hindering widespread usage of the K-ion battery technology (poor diffusion of K-ions via a solid electrode, in addition to breakdown of K post repeated cycles because of variations in volume, side reactions, dendrites development and inferior heat dissipation [208]. Fig. 15a depicts general information about potassium while Fig. 16 depicts the prospective uses of KIBs to facilitate enhanced human living [209].

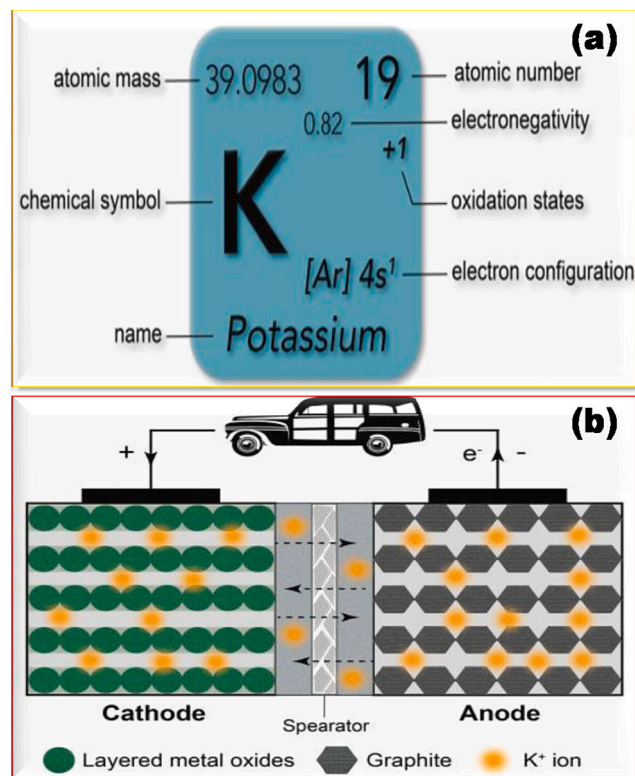


Fig. 15. (a) Basic information about K; (b) schematic elucidation of the "rocking chair" model of KIBs for power battery [209].

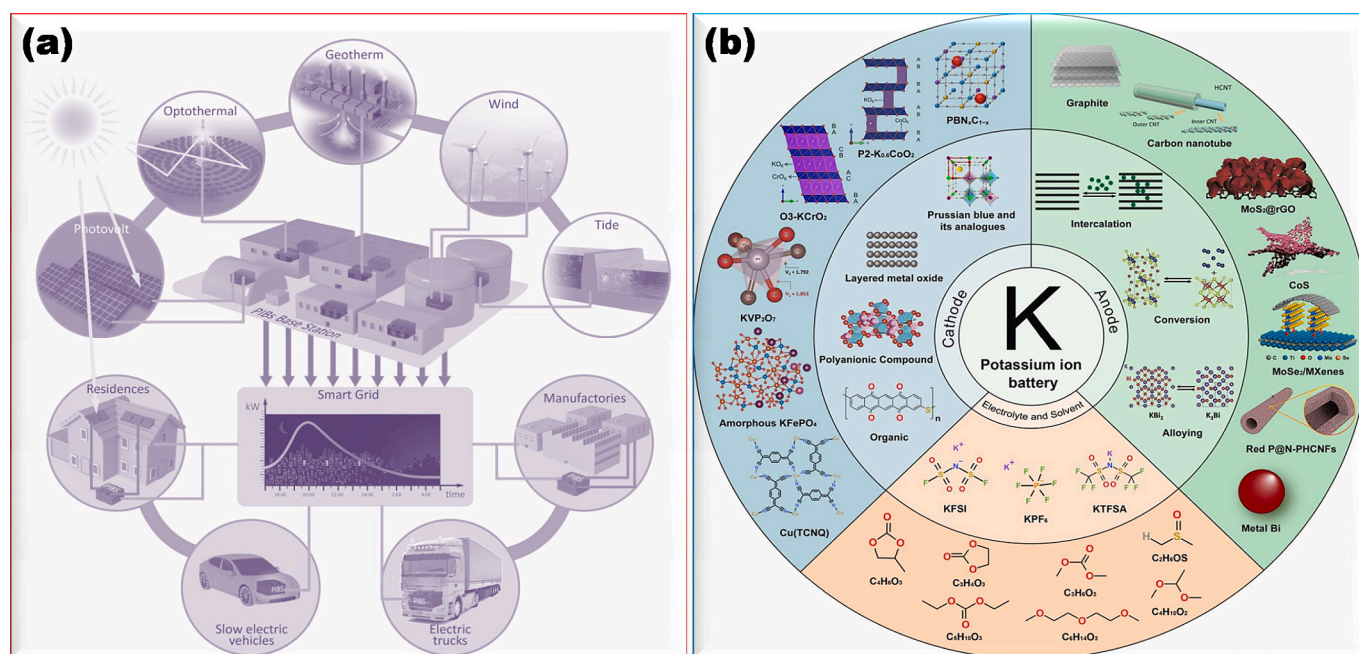


Fig. 16. (a) Elucidation of KIBs prospective uses to facilitate better future life for mankind. (b) Schematic elucidation of cathodes, anodes, electrolytes, and solvents for KIBs systems [209].

13. Conclusion, challenges and future perspectives

In this review, insight into the design, preparation, fabrication, properties and use of 2-D P—N and P—N/polymeric nanoarchitectures in electrochemical energy storage (batteries and SCs) with further insight in energy conversion have been presented. Single-layer P—N offer numerous windows of opportunities for studying fundamental and feasible applications. Elevated-quality P—N is capable of being successfully fabricated via electrochemical exfoliation, mechanical cleavage, and liquid exfoliation. A versatile range of practical and theoretical evaluations have revealed the exceptional mechanical, transport and electronic features of P—N and polymeric nanoarchitectures. These features are elevatedly anisotropic and sheet-like oriented. Hence, 2-D P—N is a potentially elevated-performing electrode substrate for numerous EESDs attributable to inherent peculiar features. Due to inherent architectural anisotropy, ease of surfacial interaction, as well as broad sheet-like spacing, P—N critically functions in electrochemical storage (LIBs, SIBs, LSBs, MIBs, KIBs and SCs), energy conversion (thermos-electronic gadgets and photovoltaic cells), as well as photocatalysts for the oxygen evolution reaction, hydrogenation, as well as hydrogen generation [209–215].

Though, numerous notable achievements have been attained with 2-D P—N, fundamental investigation and ES application researches on 2D P—N still remain in its morning stages, as significant efforts are required to mitigate challenges and evaluate emerging concepts. Thus, notable challenge is the fabrication of 2-D P—N. Due to inherently peculiar features of P—N, it presents an electrochemical behavior which is highly focused on its strategy of fabrication. A simplified strategy for tape exfoliation results in single-layer P—N with elevated quality. Nevertheless, the yield out-put of 2-D P—N through tape exfoliation is poor and not capable of large-scale manufacturing required in EESDs.

Alternatively, P—N can be garnered at a large scale via liquid-phase exfoliation, such as sonication-facilitated liquid phase exfoliation and electrochemical exfoliation. Nevertheless, liquid-phase exfoliations ultimately inculcate intrinsically and extrinsically affiliated defects in P—N, which severely affect the electrochemical features. Hence, feasible approaches should be evolved for controlling deficiencies. A remarkable bottom-up technique CVD may offer efficient strategy in fabricating

defect-free 2-D P—N or P—N displaying controllable flaws. Nevertheless, the construction of P—N by CVD technique remain challenging because of the paucity of appropriate material. Established bottom-up strategies (CVD and hydrothermal techniques), should undergo further development.

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Declaration of competing interest

No conflict is declared.

Data availability

No data was used for the research described in the article.

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